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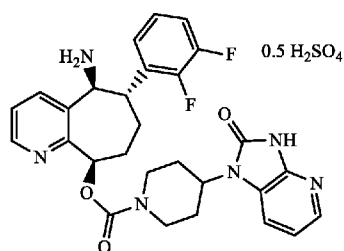
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Declarations under Rule 4.17:

- as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))
- of inventorship (Rule 4.17(iv))

(54) Title: SOLID STATE FORMS OF (5S,6S,9R)-5-AMINO-6-(2,3DIFLUOROPHENYL)-6,7,8,9-TETRAHYDRO-5H-CYCLOHEPTA[B]PYRIDIN-9-YL 4-(2-OXO-2,3-DIHYDRO-1H-IMIDAZO[4,5-B]PYRIDIN-1-YL)-1-PIPERIDINECARBOXYLATE HEMISULFATE AND PROCESSES FOR PREPARATION THEREOF



Formula-1a

(57) Abstract: The present invention relates to new solid state forms of (5S,6S,9R)-5-amino-6-(2,3-difluorophenyl)-6,7,8,9-tetrahydro-5H-cyclohepta[b]pyridin-9-yl 4-(2-oxo-2,3-dihydro-1H-imidazo[4,5-b]pyridin-1-yl)-1-piperidinecarboxylate hemisulfate represented by the following structural formula-1a which are useful for the preparation of various pharmaceutical formulations. The present invention further relates to processes for the preparation of new solid state forms of (5S,6S,9R)-5-amino-6-(2,3-difluorophenyl)-6,7,8,9-tetrahydro-5H-cyclohepta[b]pyridin-9-yl 4-(2-oxo-2,3-dihydro-1H-imidazo[4,5-b]pyridin-1-yl)-1-piperidinecarboxylate hemisulfate. Formula-1a

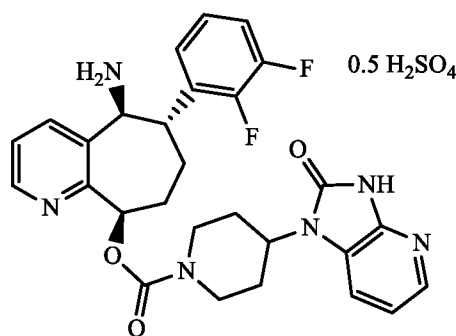
Solid state forms of (5S,6S,9R)-5-amino-6-(2,3-difluorophenyl)-6,7,8,9-tetrahydro-5H-cyclohepta[b]pyridin-9-yl 4-(2-oxo-2,3-dihydro-1H-imidazo[4,5-b]pyridin-1-yl)-1-piperidinecarboxylate hemisulfate and processes for preparation thereof

Related Applications:

This application claims the benefit of priority of our Indian patent applications 202241014804 filed on March 17, 2022 and 202241072602 filed on December 15, 2022 which are incorporated herein by reference.

Field of the Invention:

The present invention provides new solid state forms of (5S,6S,9R)-5-amino-6-(2,3-difluorophenyl)-6,7,8,9-tetrahydro-5H-cyclohepta[b]pyridin-9-yl 4-(2-oxo-2,3-dihydro-1H-imidazo[4,5-b]pyridin-1-yl)-1-piperidinecarboxylate hemisulfate represented by the following structural formula-1a and processes for preparation thereof.



Formula-1a

Background of the Invention:

(5S,6S,9R)-5-Amino-6-(2,3-difluorophenyl)-6,7,8,9-tetrahydro-5H-cyclohepta[b]pyridin-9-yl 4-(2-oxo-2,3-dihydro-1H-imidazo[4,5-b]pyridin-1-yl)-1-piperidinecarboxylate hemisulfate is commonly known as Rimegepant sulfate. It is approved by USFDA on Feb 27, 2020 and is being marketed under the brand name NURTEC ODT.

NURTEC ODT is a calcitonin gene-related peptide receptor antagonist indicated for the acute treatment of migraine with or without aura in adults and preventive treatment of episodic migraine in adults.

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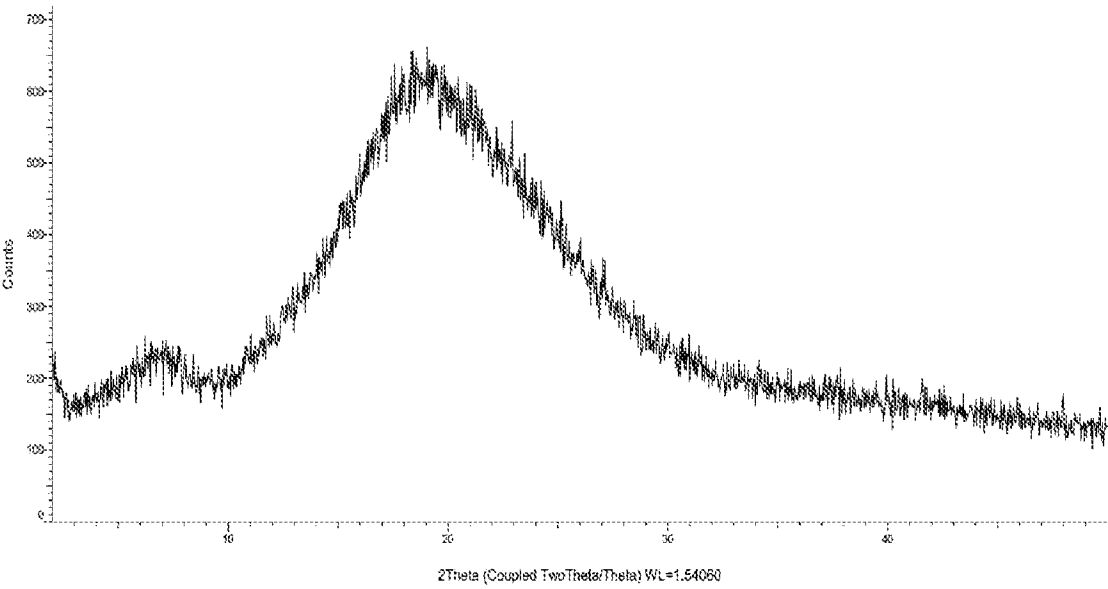


Figure-1

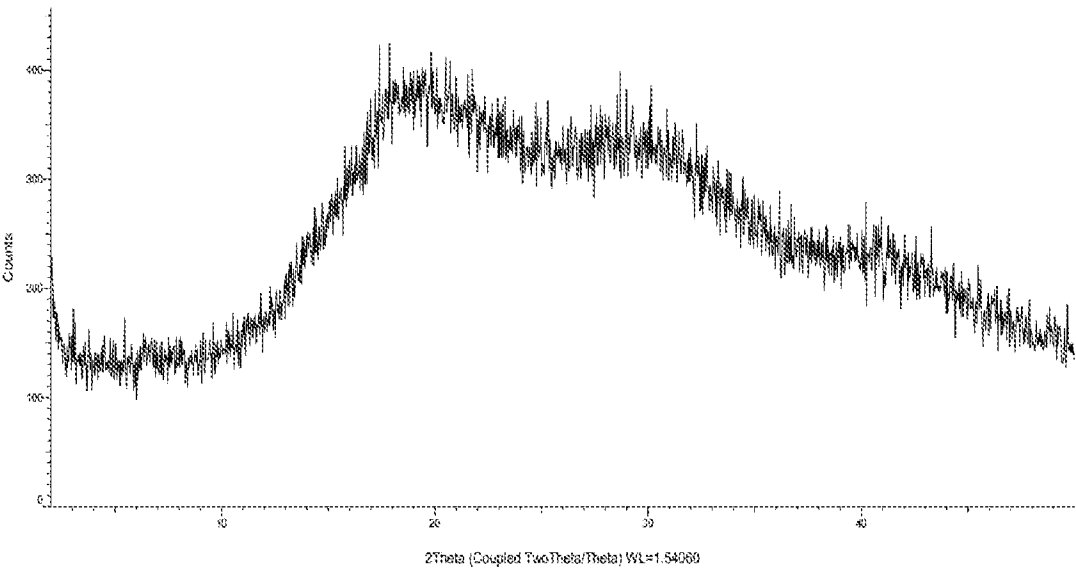
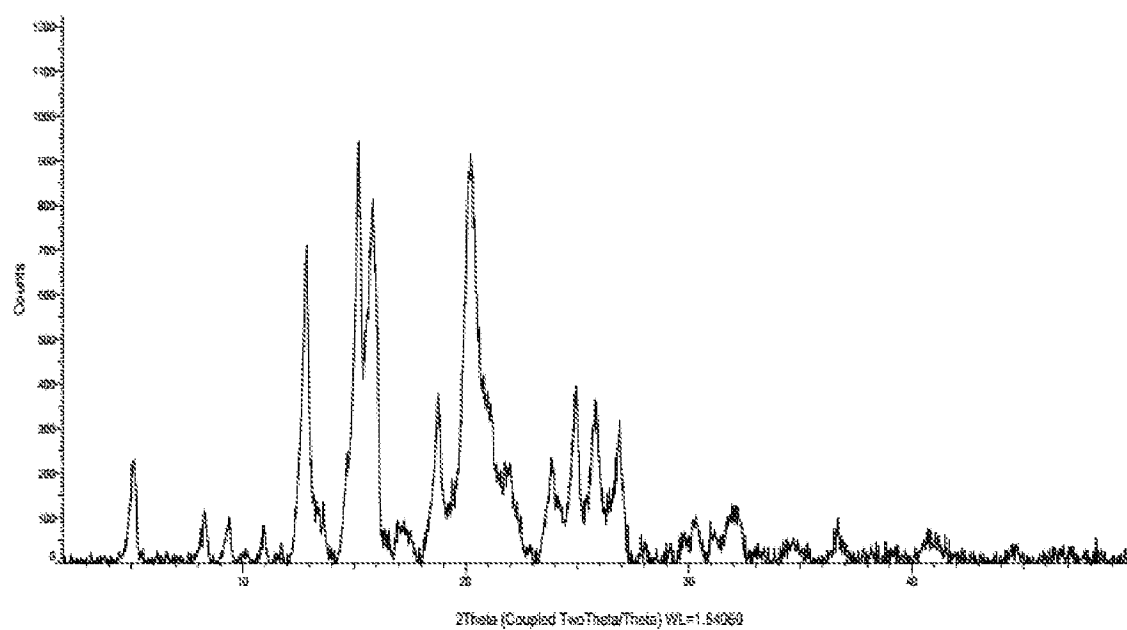
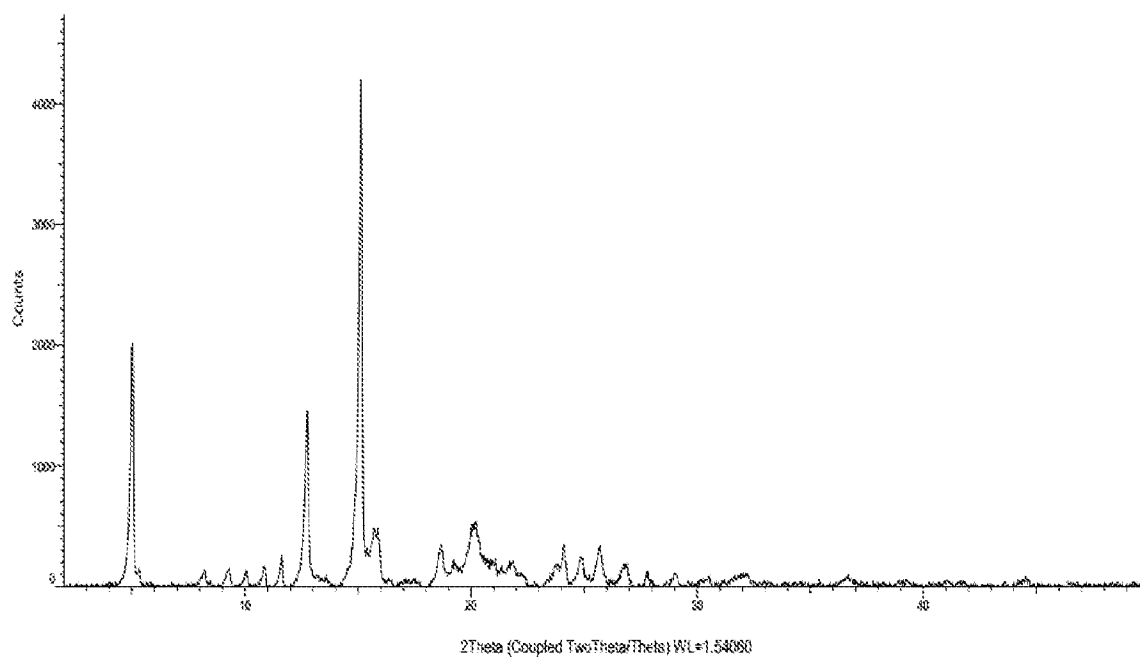


Figure-2

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**Figure-9****Figure-10**